

TRANSPORTATION TECHNOLOGY

Transportation affects our lives in a multitude of ways. Transportation systems move raw materials to manufacturers and finished products to consumers locally, nationally, and globally. Individuals use transportation systems every day for business, work, and pleasure.

Transportation technology courses provide students with opportunities to understand transportation systems from the perspective of either the consumer or the service provider. The range of courses enables students to study both vehicle ownership and vehicle maintenance, and to develop skills and prepare for careers in the servicing and repair of vehicles, aircraft, and/or watercraft. In addition, students will develop the Essential Skills and work habits that are important for success in the transportation industry.

The list of approved emphasis areas for transportation technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Transportation Technology, Grade 12

College Preparation

TTJ4C

This course enables students to further develop technical knowledge and skills as they study, test, service, and repair engine management systems; powertrains; steering/control, suspension, brake, and body systems on vehicles, aircraft, and/or watercraft; and/or small-engine products. Students will refine communication and teamwork skills through practical tasks, using a variety of tools and equipment. Students will expand their awareness of environmental and societal issues related to transportation and their knowledge of apprenticeship and college programs leading to careers in the transportation industry.

Prerequisite: Transportation Technology, Grade 11, College Preparation

A. TRANSPORTATION TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of the operation of engine management systems;
- A2.** demonstrate an understanding of the principles of operation of power transfer devices;
- A3.** demonstrate an understanding of troubleshooting techniques used to diagnose problems in powertrain systems;
- A4.** assess the effect of modifying a vehicle, aircraft, or watercraft;
- A5.** demonstrate accurate and appropriate use of technical and mathematical knowledge and skills in the study of transportation technology.

SPECIFIC EXPECTATIONS

A1. Understanding Engine Management Systems

By the end of this course, students will:

- A1.1** measure and explain the correlations among voltmeter, ammeter, and ohmmeter measurements of an electric or electronic circuit (*e.g., effect of resistance on voltage, relationship of resistance to voltage drop*);
- A1.2** identify and describe sensors, actuators, transducers, and control devices commonly used in engine management systems;
- A1.3** explain the principles of operation of various engine management systems (*e.g., emission control, multiplexing, fuel delivery management, ignition timing management*) and describe how these systems are interrelated;
- A1.4** explain how environmentally harmful gases are produced through combustion and how the engine management systems control the level of emissions in the exhaust gas (*e.g., through after-treatment of exhaust gases, exhaust gas recirculation, vapour recovery, positive crankcase ventilation, variable valve timing*);
- A1.5** explain how engine management systems may be affected by lubrication and coolant systems requiring maintenance (*e.g., a thermostat stuck open will cause the engine to run below optimal operating temperature, causing excessive fuel consumption*).

A2. Understanding Power Transfer Devices

By the end of this course, students will:

- A2.1** describe how energy is converted into motion (*e.g., chemical to mechanical, chemical to electrical, electrical to mechanical*);
- A2.2** compare the torque and power characteristics of various power sources (*e.g., two-stroke, four-stroke, gasoline, diesel, electric, fuel cell, hybrid, turboprop, jet*);
- A2.3** evaluate the performance of various power sources (*e.g., in terms of transmission output, hydraulic efficiency, volumetric efficiency, horsepower, torque*);
- A2.4** identify and trace the flow of power through the major components of a drivetrain (*e.g., transmission to axle, outboard drive, snowmobile drive, hydrostatic drive*);
- A2.5** describe power flow, gear ratios, and torque multiplication in common mechanical drive systems (*e.g., belt, chain, and gear drive systems; variable speed pulleys; planetary gear sets; clutch drives*).

A3. Troubleshooting the Powertrain

By the end of this course, students will:

- A3.1** systematically troubleshoot problems in vehicles or craft by using appropriate diagnostic

steps (e.g., *gather information, generate solutions, choose and apply a solution, validate the repair*) and equipment (e.g., *scan tool, multimeter, break-out box*);

A3.2 repair various problems in the powertrain system (e.g., *no start; problems with starting system, charging system, ignition system*);

A3.3 describe symptoms (e.g., *noise, vibration, odour, drag*) resulting from failure or improper assembly of various powertrain components (e.g., *vehicle height has dropped due to a broken coil spring, causing the driveshaft/axle working angle to exceed the manufacturer's specifications; pulleys are misaligned, causing premature belt wear*), and recommend appropriate repairs.

A4. Modifying Vehicles and/or Craft

By the end of this course, students will:

A4.1 explain the effects that body modifications (e.g., *race kits, lift kits, spoilers, ground effects*) have on vehicle or craft dynamics;

A4.2 explain the effects that vehicle or craft modifications (e.g., *changing tire size, modifying fuel and exhaust systems, changing electronic control modules [ECMs], installing lift and lower kits*) have on interrelated mechanical systems.

A5. Technological and Mathematical Literacy

By the end of this course, students will:

A5.1 demonstrate correct and appropriate use of technical terminology when preparing documentation commonly used in the transportation industry (e.g., *trouble trees, flow charts, work orders, technical service bulletins*);

A5.2 define and correctly calculate measurements related to vehicles or craft (e.g., *gear ratios, rotor run-out, engine displacement, efficiencies, output*);

A5.3 demonstrate an understanding of scientific concepts (e.g., *direct and alternating current, expansion when heat is added*) as they apply to service and repair procedures;

A5.4 prepare technical reports (e.g., *work orders, journals, parts lists*) for a variety of audiences, using appropriate language and demonstrating competent writing skills and appropriate computer literacy skills.

B. TRANSPORTATION TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** demonstrate the use of a variety of troubleshooting techniques to service and repair engine management systems;
- B2.** inspect, service, and repair drivetrain components in compliance with manufacturers' standards;
- B3.** use proper procedures for the inspection, servicing, and repair of steering/control, suspension, brake, and body systems;
- B4.** develop appropriate solutions to a variety of repair challenges.

SPECIFIC EXPECTATIONS

B1. Troubleshooting Engine Management Systems

By the end of this course, students will:

- B1.1** access and correctly interpret data provided by the on-board diagnostic system (*e.g., dash codes, on-board diagnostic [OBD] data*);
- B1.2** use trouble charts and manufacturers' diagnostic procedures correctly and appropriately to service problems indicated by on-board diagnostic data;
- B1.3** use appropriate equipment to diagnose and repair engine control systems and components (*e.g., sensors, actuators, control devices*) according to manufacturers' recommendations;
- B1.4** demonstrate proper use of diagnostic equipment to avoid damage to equipment and/or vehicle or craft components (*e.g., damage caused by a short circuit, a voltage spike, an ammeter connected in series*).

B2. Service and Repair of Drivetrain Components

By the end of this course, students will:

- B2.1** inspect and diagnose problems in drivetrain components (*e.g., transmission, clutch, driveshaft, outboard drive*) in compliance with manufacturers' standards;

- B2.2** perform service procedures on drivetrain components (*e.g., replace and adjust clutch, replace universal joints, replace/repack bearings, replace drive seals, adjust belt or chain*) safely and correctly.

B3. Service and Repair of Steering/Control, Suspension, Brake, and Body Systems

By the end of this course, students will:

- B3.1** inspect and measure component tolerances when servicing steering/control, suspension, and brake systems and body component alignment (*e.g., steering/control systems: tire wear, ball joint play; suspension systems: ride height/trim height; brake systems: disc thickness, drum diameter; body alignment: door opening*), and compare the results with manufacturers' specifications;
- B3.2** demonstrate proper procedures and the safe use of specialty tools and equipment in the service and repair of steering/control, suspension, brake, and body system components (*e.g., steering/control: tie rod fork, tire balancer; suspension: coil spring compressor; brakes: lathe, dial indicator gauge; body: metal inert gas [MIG] welder*).

B4. Solving Repair Challenges

By the end of this course, students will:

- B4.1** access and use appropriate resources (*e.g., repair manuals, online resources, equipment instructions*) as required to successfully address repair challenges;
- B4.2** demonstrate the safe and correct use of a variety of soldering, heating, cutting, and/or welding equipment for service repair and modification tasks;

- B4.3** perform an inspection for various repair challenges (*e.g., brake service, ball-joint replacement, driveshaft or belt misalignment, transmission repair*) and prepare a report on work to be done, including a cost-benefit analysis.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of environmental issues in the transportation industry, and use best practices to remedy or reduce the environmental effects of using specific products or processes;
- C2.** assess the effects that various aspects of the transportation industry have on society.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** demonstrate an understanding of ways in which the transportation industry affects the environment and of efforts being made to remedy or reduce harmful effects (*e.g., improved production methods, automotive parts recycling*), including ways of disposing of waste products (*e.g., used oil, used batteries, used paints/thinners*);
- C1.2** describe appropriate actions to be taken in the event of a spill of waste products (*e.g., gasoline, antifreeze*) and demonstrate the ability to safely implement such actions (*e.g., implement an emergency action plan to contain and clean up the spill*);
- C1.3** identify the procedures required to prevent the release of ozone-depleting materials and other harmful substances (*e.g., electrolyte, antifreeze, gasoline*) during the servicing of vehicle or craft systems.

C2. Technology and Society

By the end of this course, students will:

- C2.1** assess the economic effects of the transportation industry on the local and global levels;
- C2.2** assess the importance to society of a transportation infrastructure (*e.g., highway system, rail system, canal system*);
- C2.3** evaluate the potential benefits (*e.g., technical, safety, financial, business*) to society of emerging technologies related to the transportation industry (*e.g., collision avoidance system, run-flat tires, fuel cells, composite materials*);
- C2.4** assess the pros and cons of various types of fuel/energy sources (*e.g., gasoline, propane, diesel, electrical/battery power, biodiesel, hybrid powerplant, hydrogen power cells*) used to power vehicles or small-engine products, taking into account a variety of perspectives (*e.g., consumer's perspective: cost to purchase, cost to operate, performance, emission levels; service/repair perspective: training, safety issues, new tools/equipment required*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate the use of professional work practices and procedures and compliance with occupational health and safety regulations and standards;
- D2.** describe career opportunities in the transportation industry and the training and qualifications required for them.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** demonstrate an understanding of legislation and regulations related to procedures and operations used in transportation technology facilities (*e.g., Occupational Health and Safety Act [OHSA]; regulations and standards outlined in the Workplace Hazardous Materials Information System [WHMIS]; Apprenticeship and Certification Act [ACA]*);
- D1.2** demonstrate good housekeeping and safety practices in the work environment (*e.g., cleaning up spills and leaks, keeping areas clean and clear of obstructions*);
- D1.3** identify potential health risks (*e.g., asbestos dust, fumes from brake fluid and cleaners*) when servicing vehicles or craft, and demonstrate the use of safe procedures to mitigate these hazards (*e.g., use appropriate ventilation and breathing protection*);
- D1.4** use protective clothing and equipment (*e.g., eye and hearing protection, gloves, breathing apparatus, hoist, safety stand*) as required to ensure their own and others' safety in the work environment;
- D1.5** explain the need for and demonstrate the ability to follow appropriate safety precautions applicable to new technologies when working around or servicing vehicles, craft, or power equipment (*e.g., precautions regarding high current and voltage, capacitor discharge rate of supplemental restraint systems, extreme temperature of exhaust systems, accumulators in hydraulic systems, pressure vessels*);
- D1.6** demonstrate an understanding of professional responsibilities in the transportation industry with regard to personal and public safety (*e.g.,*

good workmanship, integrity, high-quality customer service, compliance with manufacturers' standards).

D2. Career Opportunities

By the end of this course, students will:

- D2.1** describe on the basis of research the qualifications required for careers in the transportation industry (*e.g., apprenticeship standards, certifications, licensing requirements*);
- D2.2** identify the regulations regarding restricted skill sets in the Apprenticeship and Certification Act (available at www.e-laws.gov.on.ca);
- D2.3** demonstrate an understanding of the need for continuous learning and skill upgrading created by emerging technological developments (*e.g., high voltages in hybrid vehicles, alternative fuels, new types of refrigerant*) and the changing repair techniques required for modern vehicles and craft;
- D2.4** demonstrate an understanding of and apply the Essential Skills that are important for success in the transportation industry, as identified in the Ontario Skills Passport (*e.g., decision making, problem solving, finding information*);
- D2.5** demonstrate an understanding of and apply the work habits that are important for success in the transportation industry, as identified in the Ontario Skills Passport (*e.g., teamwork, reliability, initiative*);
- D2.6** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in transportation technology (*e.g., Passport to Safety certificate, skills checklist, Ontario Skills Passport Work Plan and Transition Plan*), and explain why having a current portfolio is important for career development and advancement.